

THE DI-HYDRO VISUALIZATION TOOL

- USER MANUAL -

Developed by **Accent Pro 2000 S.R.L. (AP2K)**
for the **Di-Hydro Project**



0. Running the Application

0.1 Installing from GitHub

To install the Visualization Tool from GitHub, run in the terminal:

git clone <https://github.com/TheoN03/The-Di-Hydro-Visualization-Tool>

0.2 Running it locally

[1] To run **The Visualization Tool**, the users need to go to the **main folder: The Di-Hydro Visualization Tool\Setup Files\di_hydro_installer_pack\dist\The Di-Hydro Visualization Tool\The Di-Hydro Visualization Tool.exe** and double click it.

[2] If the **.exe** file throws an error, try reinstalling it by double clicking **build_installer.bat**, that is here: **The Di-Hydro Visualization Tool\Setup Files\di_hydro_installer_pack**

After the installation, rerun **The Di-Hydro Visualization Tool.exe**, located in the same place as step [1].

1. TAB 1 – PPC DATA DISPLAY



Figure 1 TAB 1 - PPC DATA

1.1 Data Loading and Visualization

The first tab is designed for quick and straightforward use. To begin, simply load your data file into the application. The Visualization Tool supports multiple data formats, including **.xlsx**, **.csv**, and **.json**.

Once the data has been successfully loaded, all available parameters will be automatically detected and displayed. Each parameter is presented with its corresponding name and historical values, which are visualized on an interactive graph for easier analysis and monitoring.

The figure below shows a zoomed-in example of a parameter and its historical trend visualization.

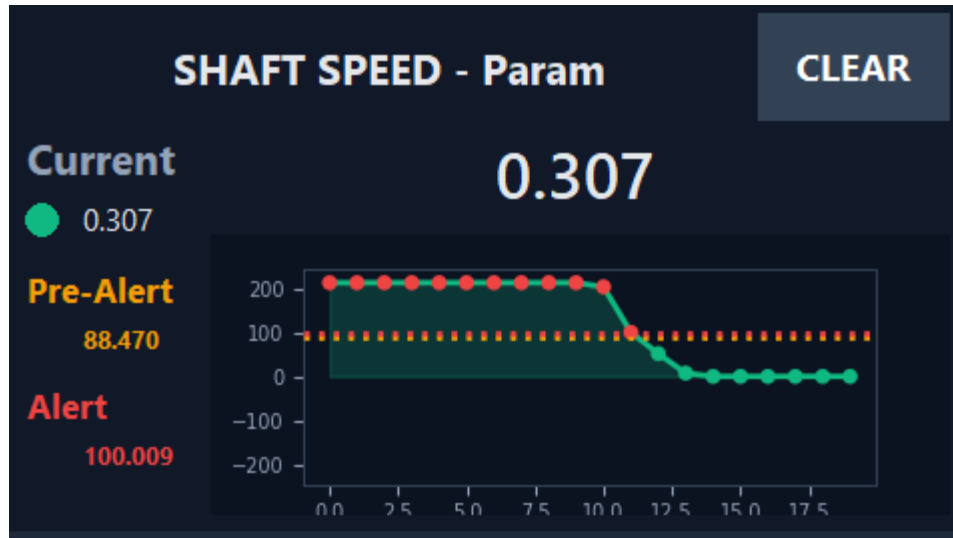


Figure 2 TAB 1 - Single Parameter Section

As shown in the figure, each parameter section provides the following information and controls:

- **Current Value** - Displays the latest measured value of the parameter.
- **Pre-Alert Limits** - Indicates the threshold values that trigger a warning condition before reaching a critical state.
- **Alert Limits** - Defines the critical threshold values that generate an alarm condition.
- **Historical Trend Graph** - Visualizes the parameter's historical data for monitoring and analysis.
- **Clear Button** - Allows the operator to acknowledge and clear active warnings or alarms.

Most functions within **Tab 1** are fully automated. Once data has been loaded, the system automatically updates parameter values, evaluates alert conditions, and refreshes the graphical displays in real time.

The only operator actions required are:

- Loading the data source
- Starting the data stream
- Pausing the data stream
- Stopping the data stream
- Clearing acknowledged warnings and alarms using the **CLEAR** button
- Selecting the HPP component to display

This automation minimizes operator intervention while ensuring continuous monitoring and timely notification of abnormal conditions.

The figure below shows the buttons for different Hydro Power Plant components:

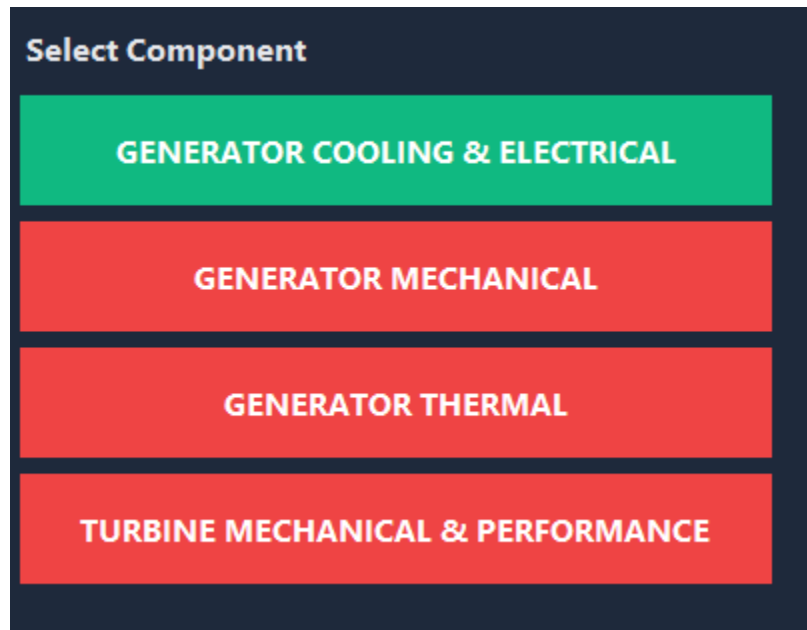


Figure 3 TAB 1 - Hydro Power Plant Component Selection

Operators can press these buttons to change the displayed parameter collection.

2. TAB 2 – AUGMENTED REALITY (AR)

2.1. Main Window & Layout Guide

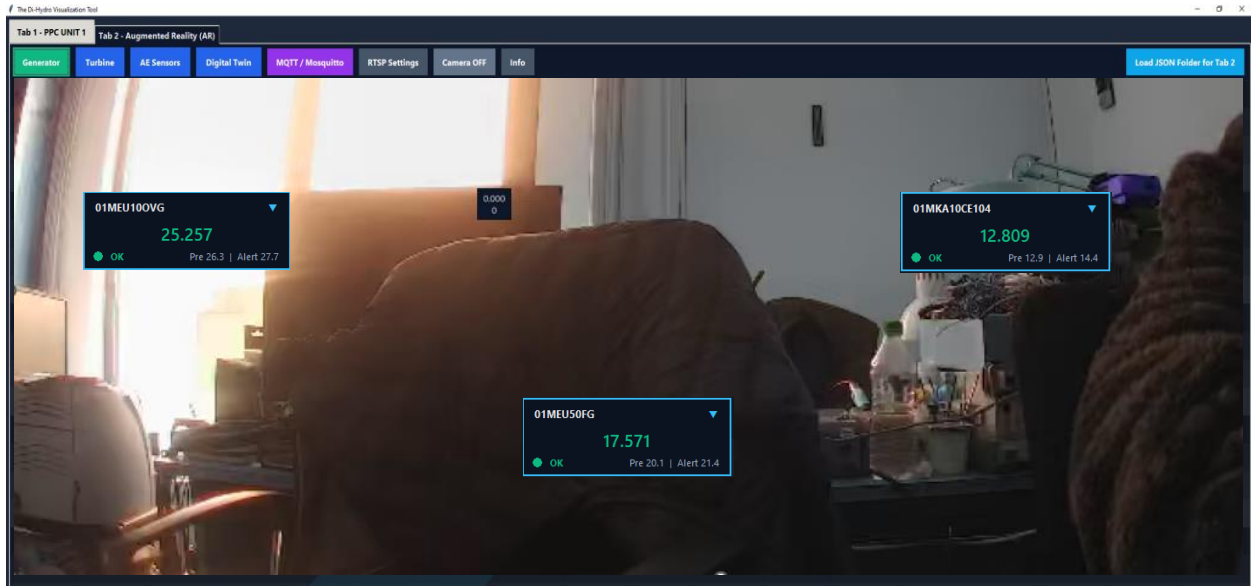


Figure 4 TAB 2 - Augmented Reality Window

The second tab of The Visualization Tool provides an Augmented Reality (AR) interface for monitoring various hardware components and operational parameters. Each hardware category is associated with its own dedicated parameter set and live camera stream background, allowing operators to visualize real-world equipment alongside relevant data.

In the example shown above, three key generator parameters are displayed together with their corresponding status indicators, enabling rapid assessment of the generator's operating condition.

To use the Augmented Reality visualization, operators must first load the required configuration data file (*.json*). Once the data has been successfully loaded, the desired hardware category can be selected using the category selection buttons shown below. The displayed parameters and camera stream will automatically update according to the selected category.

2.2. Utility Buttons Section



Figure 5 TAB 2 - Utility Buttons

The category selection buttons allow operators to switch between the available hardware categories, updating both the displayed parameters and the corresponding camera stream in the Augmented Reality tab.

Additional controls are provided to enhance usability and system configuration:

- **Info** - Opens a brief tutorial containing guidance on the operation of the Augmented Reality interface.
- **Camera ON/OFF** - Starts or stops the live camera stream displayed in the background.
- **Configuration Buttons** - Two additional configuration buttons are available for advanced settings and customization. Their functionality is described in detail in the following sections of this manual.

These controls enable operators to customize the displayed information and manage the visualization environment according to operational requirements.

2.2.1. MQTT/MOSQUITTO Button

This button opens a configuration window for the Mosquitto broker.

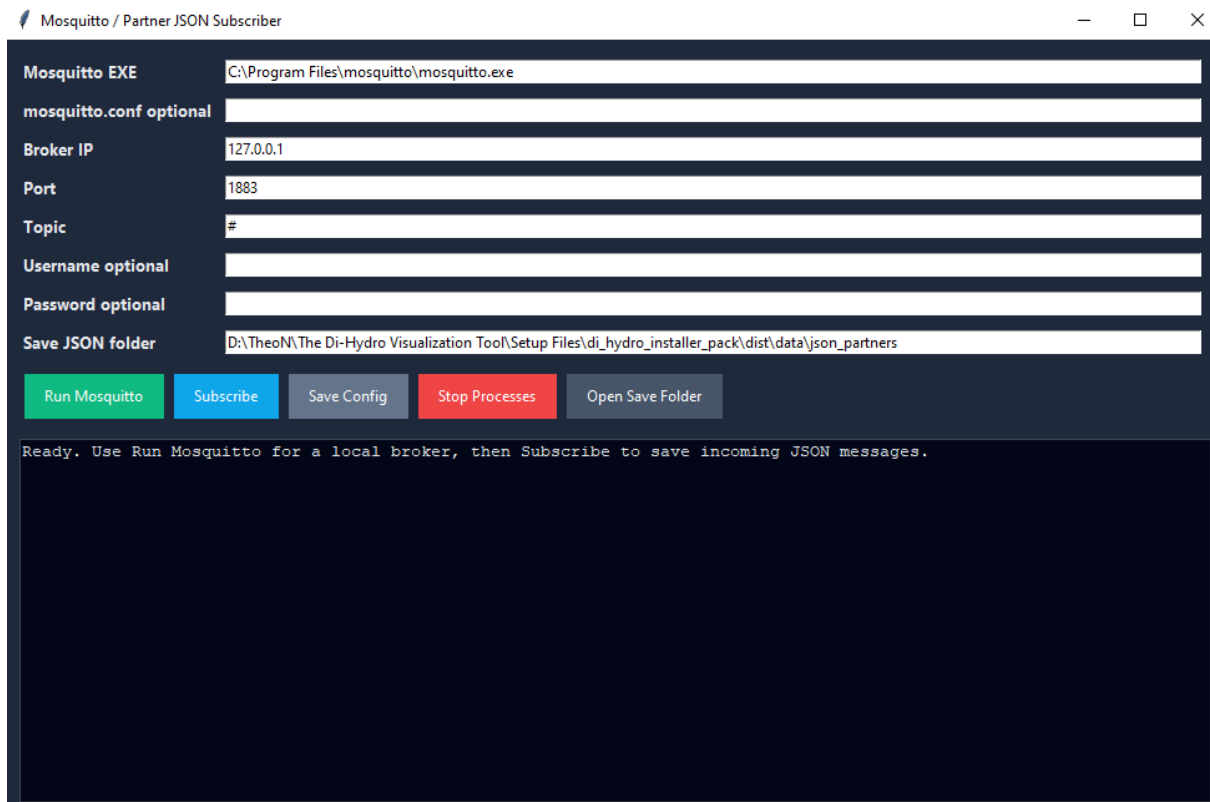
The **MQTT Configuration Window** provides a graphical interface for managing MQTT broker connections and subscriptions. Its functionality is equivalent to running the MQTT broker and subscription commands through a PowerShell terminal, however, all operations can be performed directly through dedicated buttons within the application.

Using this interface, operators can:

- Connect to and start communication with the MQTT broker.
- Subscribe to the required MQTT topics.
- Save broker credentials and configuration settings.
- Store and manage received data.

The MQTT Configuration Window can operate in parallel with the Visualization Tool, allowing data acquisition and visualization to run simultaneously without interrupting system monitoring activities.

An example of the MQTT Configuration Window is shown below.



The screenshot shows a software window titled "Mosquitto / Partner JSON Subscriber". It features a dark-themed interface with several configuration fields and a row of action buttons. The fields are labeled on the left and have corresponding input boxes on the right. The fields and their values are: "Mosquitto EXE" (C:\Program Files\mosquitto\mosquitto.exe), "mosquitto.conf optional" (empty), "Broker IP" (127.0.0.1), "Port" (1883), "Topic" (#), "Username optional" (empty), "Password optional" (empty), and "Save JSON folder" (D:\TheoN\The Di-Hydro Visualization Tool\Setup Files\di_hydro_installer_pack\dist\data\json_partners). Below the fields is a row of five buttons: "Run Mosquitto" (green), "Subscribe" (blue), "Save Config" (grey), "Stop Processes" (red), and "Open Save Folder" (grey). At the bottom of the window is a large text area containing the message: "Ready. Use Run Mosquitto for a local broker, then Subscribe to save incoming JSON messages."

Figure 6 TAB 2 - MQTT Configuration Window

For learning how to install and use the Mosquitto broker, please, read the documentation and tutorials provided by **IMT's Data Exchange Platform**:

<https://github.com/satrai-lab/di-hydro-platform>

2.2.2. RTSP Settings Button

RTSP (Real-Time Streaming Protocol) is a network communication protocol used to establish, control, and manage real-time audio and video streams over IP networks. It enables functions such as starting, stopping, pausing, and monitoring live video feeds from compatible devices.

To utilize the camera streaming functionality within the Visualization Tool, operators must use cameras that support RTSP communication. The **RTSP Settings** button opens the RTSP Configuration Window, which allows operators to configure the connection parameters required to access a camera stream.

Within this configuration window, operators can enter:

- The camera IP address.
- RTSP connection details.
- Authentication credentials (username and password).
- Additional connection parameters, if required by the camera device.

This configuration mechanism provides flexibility when connecting to cameras deployed on different networks or using different IP addresses, allowing operators to quickly update connection settings without modifying the application itself.

NOTICE! Please, keep in mind that not all cameras let's end-users activate or deactivate the RTSP settings! Do your research when buying a camera, so you know it can work with remote connection.

From our experience and tests, the cameras from Ezviz work.

An example of the **RTSP Configuration Window** is shown below.

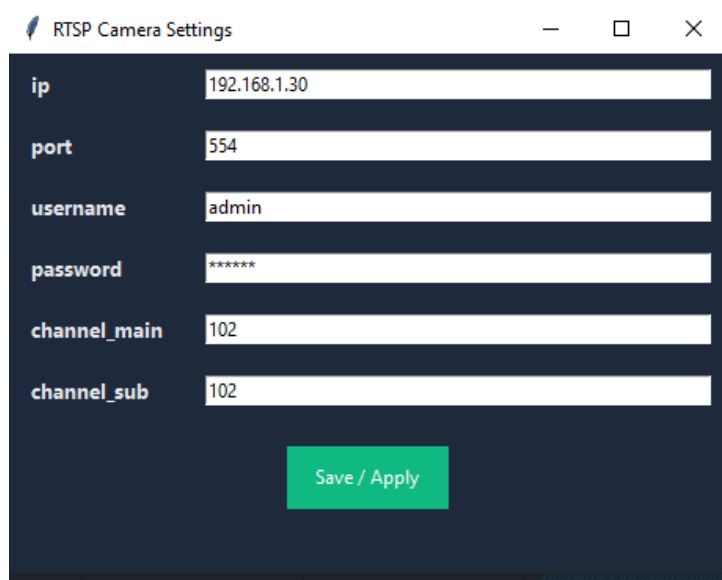
The image shows a software window titled "RTSP Camera Settings". It contains several input fields for configuration: "ip" with the value "192.168.1.30", "port" with "554", "username" with "admin", "password" with "*****", "channel_main" with "102", and "channel_sub" with "102". At the bottom of the window is a green button labeled "Save / Apply". The window has standard OS window controls (minimize, maximize, close) in the top right corner.

Figure 7 TAB 2 - RTSP Settings Configuration Window

3. Manual Summary & Additional Information

3.1 Document Control

Item	Description
Project	DI-HYDRO
Document Type	User Manual
Version	1.0
Developer	Accent Pro 2000 S.R.L. (AP2K)
Audience	HPP Operators, Engineers, Administrators

3.2 Revision History

Version	Date	Author	Description
1.0	June 2026	AP2K	Initial Release

3.3 Steps

3.3.1 Introduction

The DI-HYDRO Visualization Tool is a monitoring and visualization application developed within the DI-HYDRO project. The software enables Hydro Power Plant (HPP) operators to monitor

operational parameters, visualize historical data, and access Augmented Reality (AR) displays of plant equipment using live camera feeds.

3.3.2. System Requirements

Component	Requirement
Operating System	Windows 10/11
RAM	8 GB or higher
Storage	500 MB free space
Network	Local network or Internet access
Camera	RTSP-compatible IP camera (for AR mode)

3.3.3 Installation and Startup

Clone the repository or run the provided executable package. If startup fails, execute **build_installer.bat** and relaunch the application.

3.3.4 Tab 1 - PPC Data Display

Load **.xlsx, .csv, or .json files**. Parameters are automatically detected and displayed with current values, alert thresholds, and historical trend graphs.

3.3.5 Tab 2 - Augmented Reality (AR)

Provides live equipment visualization with camera streams and parameter overlays. Operators load a JSON configuration file and select hardware categories for display.

3.3.6 MQTT Configuration

Allows operators to connect to **MQTT brokers**, subscribe to topics, save credentials, and manage incoming data through a graphical interface.

3.3.7 RTSP Configuration

Allows configuration of **RTSP-enabled cameras** through **IP address, credentials, and connection parameters**.

3.3.8 Troubleshooting

Problem	Possible Cause	Solution
No parameters displayed	Data file not loaded	Load a valid data file
Camera stream unavailable	RTSP disabled	Enable RTSP and verify credentials
MQTT connection failed	Invalid broker settings	Verify broker address and credentials

3.3.9 Abbreviations

Abbreviation	Meaning
AR	Augmented Reality
HPP	Hydro Power Plant
MQTT	Message Queuing Telemetry Transport
RTSP	Real-Time Streaming Protocol
PPC	Power Plant Controller